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Fuel supply system and fuel supply method for aircraft gas turbine engine

Abstract

A fuel supply system includes a gear pump, and a minimum pressurizing check valve including a cylinder and a piston that partitions the internal space of the cylinder into a first chamber that communicates with the metering section and a second chamber that communicates with a system inlet, the piston is spring-biased in a direction toward the first chamber, a side wall of the cylinder is provided with an outflow port having an opening degree adjusted as the piston moves back and forth, the minimum pressurizing check valve keeps a differential pressure between the system inlet and the outlet of the metering section constant when the rotation speed of the gear pump is equal to or more than a lower threshold and is equal to or less than an intermediate threshold.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a fuel supply system for an aircraft gas turbine engine, and particularly to a fuel supply system configured to control a fuel flow rate by controlling a rotation speed of an electric motor that drives a pump, the fuel supply system being configured to control the rotation speed of the electric motor such that a deviation between an estimated value and a target value becomes zero by identifying an internal leak model of the pump during a low fuel flow rate and taking the amount of internal leakage of the pump estimated based on the model into consideration during a high fuel flow rate. Further, the present disclosure relates to a fuel supply method using the fuel supply system having such a configuration.

BACKGROUND ART

(2) A fuel supply system for an aircraft gas turbine engine is a system that supplies fuel from a fuel tank installed, for example, in a main wing of an aircraft, to a fuel nozzle installed in a combustor of a gas turbine engine (for example, a turbofan engine) mounted on the aircraft, at a predetermined flow rate depending on operating conditions of the gas turbine engine.

(3) The fuel supply system includes a pump used to increase pressure of the fuel, but the pump normally includes a low-pressure pump configured as a non-positive displacement pump (for example, a centrifugal pump) and a high-pressure pump configured as a positive displacement pump (for example, a gear pump).

(4) These pumps have conventionally been driven by power extracted from a main shaft of the gas turbine engine through an AGB (Accessory Gear Box). However, in recent years, electrification of aircraft gas turbine engines has been actively promoted to prevent global warming, and a fuel supply system configured to drive the above-described pump with an electric motor is being studied.

(5) Patent Document 1 discloses a fuel supply system for an aircraft gas turbine engine configured to control a fuel flow rate by controlling a rotation speed of an electric motor that drives a pump.

(6) The fuel supply system includes an orifice flowmeter on a downstream side of the high-pressure pump, and controls the rotation speed of the electric motor, which drives the pump, such that the deviation between the value of the fuel flow rate measured by the orifice flowmeter and the target value becomes zero during the low fuel flow rate. However, since a front-back differential pressure of the orifice flowmeter rapidly increases in proportion to the square of the fuel flow rate with the increase of the fuel flow rate, a large pump with a very large boosting capability is required when the orifice flowmeter is to be used to measure the fuel flow rate over the entire range of fuel flow rate. This results in an increase in weight, and thus it is undesirable for a fuel supply system for an aircraft gas turbine engine requiring weight reduction.

(7) Therefore, the fuel supply system is configured such that some parts of fuel flows bypassing the orifice flowmeter during the high fuel flow rate. In this state, the fuel flow rate is estimated based on the rotation speed of the high-pressure pump, and the rotation speed of the electric motor for driving the pump is controlled such that the deviation between the estimated value and the target value becomes zero.

(8) However, the high-pressure pump serving as a component of the fuel supply system is configured as a gear pump, and internal leakage gradually increase with a deterioration over time. For this reason, in order to control the fuel flow rate with high accuracy during the high fuel flow rate, it is necessary to calculate the estimated value of the fuel flow rate by taking the internal leakage into consideration.

(9) Accordingly, in the fuel supply system disclosed in Patent Document 1, identification of the internal leak model of the high-pressure pump is performed using the value of the fuel flow rate measured by the orifice flowmeter, during the low fuel flow rate at which a ratio of the amount of internal leakage to a discharge flow rate is high. The amount of internal leakage estimated based on the internal leak model is taken into consideration when the estimated value of the fuel flow rate is

calculated based on the rotation speed of the high-pressure pump during the high fuel flow rate.

RELATED ART DOCUMENT

Patent Document

(10) Patent Document 1: International Publication No. WO 2019-172372

SUMMARY OF THE DISCLOSURE

Problems to be Solved by the Disclosure

(11) An object of the present disclosure is to provide a fuel supply system for an aircraft gas turbine engine, which is formed to control a fuel flow rate by controlling a rotation speed of an electric motor for driving a pump, and to perform identification of an internal leak model of a high-pressure pump with high accuracy in a configuration different from that of Patent Document 1. Further, the present disclosure is to provide a fuel supply method using the fuel supply system having such a configuration.

Means for Solving the Problems

(12) In order to solve the above problems, a first aspect of the present disclosure is directed to a fuel supply system including: a gear pump that increases a pressure of fuel flowing in from a system inlet and discharges the fuel; a metering section that is disposed downstream of the gear pump and includes an orifice and a check valve connected in parallel with each other; a minimum pressurizing check valve that is disposed between the metering section and a system outlet; and a controller, wherein the check valve closes when a rotation speed of the gear pump is less than an upper threshold, and opens when the rotation speed of the gear pump is equal to or more than the upper threshold, the minimum pressurizing check valve includes a cylinder and a piston capable of being movable back and forth in the cylinder, the cylinder has an internal space partitioned by the piston into a first chamber that communicates with an outlet of the metering section through an inflow port and a second chamber that communicates with the system inlet, the piston receiving a force in a direction toward the first chamber from a spring disposed in the second chamber, a side wall of the cylinder is provided with an outflow port that is an opening through which the inflow port and the system outlet communicate with each other, an opening degree of the outflow port is adjusted as the piston moves back and forth, and the minimum pressurizing check valve keeps a differential pressure between the system inlet and the outlet of the metering section constant when the rotation speed of the gear pump is equal to or more than a lower threshold and is equal to or less than an intermediate threshold, the lower threshold being smaller than the upper threshold, the intermediate threshold being more than the lower threshold and smaller than the upper threshold.

Effects of the Disclosure

(13) According to a fuel supply system of the present disclosure, it is possible to identify an internal leak model with high accuracy, which is an excellent effect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram showing a configuration of a fuel supply system for an aircraft gas turbine engine according to an embodiment of the present disclosure.

(2) FIG. 2 is a schematic explanatory diagram showing a structure of a minimum pressurizing check valve provided in the fuel supply system for the aircraft gas turbine engine according to the embodiment of the present disclosure.

(3) FIG. 3A is a schematic explanatory diagram showing a mode of movement of a piston of the minimum pressurizing check valve provided in the fuel supply system for the aircraft gas turbine engine according to the embodiment of the present disclosure.

(4) FIG. 3B is a schematic explanatory diagram showing a mode of movement of the piston of the minimum pressurizing check valve provided in the fuel supply system for the aircraft gas turbine

engine according to the embodiment of the present disclosure.

(5) FIG. 3C is a schematic explanatory diagram showing a mode of movement of the piston of the minimum pressurizing check valve provided in the fuel supply system for the aircraft gas turbine engine according to the embodiment of the present disclosure.

(6) FIG. 3D is a schematic explanatory diagram showing a mode of movement of the piston of the minimum pressurizing check valve provided in the fuel supply system for the aircraft gas turbine engine according to the embodiment of the present disclosure.

MODE FOR CARRYING OUT THE DISCLOSURE

(7) Hereinafter, an embodiment of the present disclosure will be described in detail with reference to the drawings.

(8) FIG. 1 is a block diagram showing a configuration of a fuel supply system for an aircraft gas turbine engine according to an embodiment of the present disclosure.

(9) A fuel supply system **100** is a system for supplying fuel, which is supplied through a boost pump **220** from a fuel tank **210** provided in, for example, a main wing of an aircraft **200**, to a fuel nozzle **320** installed in a combustor **310** of a gas turbine engine **300** (for example, a turbofan engine) at a predetermined flow rate depending on operating conditions of the gas turbine engine **300**.

(10) The fuel supply system **100** includes, as main components, a low-pressure pump **110**, a high-pressure pump **120**, an electric motor **130**, a metering section **160** including an orifice **140** and a pressurizing valve **150**, a minimum pressurizing check valve (MPCV) **170**, and a fuel supply controller (controller) **180**. In addition to these components, the fuel supply system **100** conventionally includes a fuel filter, a fuel cooled oil cooler (FCOC), and the like, which are not directly related to the present disclosure and thus not shown.

(11) The low-pressure pump **110** is a pump that increases pressure of fuel supplied from the fuel tank **210** of the aircraft **200** through the boost pump **220** and feeds the fuel to the high-pressure pump **120**. The low-pressure pump **110** is configured as a non-positive displacement pump (for example, a centrifugal pump), and is driven by the electric motor **130**.

(12) The high-pressure pump **120** is a pump that increases pressure of the fuel supplied from the low-pressure pump **110** and feeds the fuel to a downstream side. The high-pressure pump **120** is configured as a positive displacement pump (for example, a gear pump), and is driven by the electric motor **130**. A temperature sensor ST is provided at a point P.sub.L on a fuel flow passage between the low-pressure pump **110** and the high-pressure pump **120**, and a signal of a fuel temperature T at an inlet of the high-pressure pump **120** measured by the temperature sensor ST is transmitted to the fuel supply controller **180**.

(13) Power may be transferred from the electric motor **130** to the low-pressure pump **110** and the high-pressure pump **120** in any manner. Further, the low-pressure pump **110** and the high-pressure pump **120** may be driven by individual electric motors, respectively, instead of being driven by one common electric motor **130**.

(14) The electric motor **130** has the function of driving the low-pressure pump **110** and the high-pressure pump **120** as described above. A rotation speed sensor SN is attached to the electric motor **130**, and a signal of a rotation speed N.sub.m measured by the rotation speed sensor SN is transmitted to the fuel supply controller **180**. Further, the rotation speed of the electric motor **130** is controlled by a rotation speed command signal N.sub.r output from the fuel supply controller **180**.

(15) A branch (point P.sub.H) is provided downstream of the high-pressure pump **120**, and the metering section **160**, which includes the orifice **140** and the pressurizing valve **150** connected in parallel with each other, is disposed downstream of the branch.

(16) The orifice **140** forms an orifice flowmeter together with a differential pressure gauge SDP connected in parallel with the orifice **140** (the flowmeter being used to measure a flow rate based on a flow passage (parallel flow passage) in which the orifice **140** and the differential pressure gauge SDP are disposed, which are connected in parallel with each other. The parallel flow passage

being provided in parallel with the pressurizing valve **150** (check valve) through the above-described branch). As will be described below, the orifice flowmeter has a function of measuring a fuel flow rate in a state where the pressurizing valve **150** connected in parallel with the orifice **140** is closed, and for this purpose, a signal of a front-back differential pressure $\Delta P_{sub.M}$ of the orifice **140** measured by the differential pressure gauge SDP is transmitted to the fuel supply controller **180**.

(17) When the fuel flow rate flowing into the metering section **160** is low (during a low fuel flow rate), the pressurizing valve **150** is in a closed state. In this state, all of the fuel flowing into the metering section **160** passes through the orifice **140**, and the fuel flow rate is measured (metered) by the orifice flowmeter including the orifice **140** and the differential pressure gauge SDP as described above.

(18) On the other hand, when the fuel flow rate flowing into the metering section **160** increases, the front-back differential pressure of the orifice **140** also increases, and when the differential pressure exceeds a threshold inherent to the pressurizing valve **150** which is a check valve, the pressurizing valve **150** is in an open state. In this state, a total fuel flow rate supplied from the fuel supply system **100** to the fuel nozzle **320** of the gas turbine engine **300** cannot be measured by the orifice flowmeter described above. Therefore, in this state, the total fuel flow rate is estimated based on the rotation speed of the electric motor **130** that drives the high-pressure pump **120**, as will be described below.

(19) The minimum pressurizing check valve **170** is provided downstream of the parallel flow passage described above, and is a valve having a function of keeping a differential pressure $\Delta P_{sub.T}$ (see FIG. 1) between an inlet (point $P_{sub.I}$) of the fuel supply system **100** and an outlet (point $P_{sub.M}$) of the metering section **160** constant during a low fuel flow rate at which identification of an internal leak model of the high-pressure pump **120** is performed, as will be described below, but a structure thereof will first be described below.

(20) FIG. 2 is a schematic explanatory diagram showing a structure of the minimum pressurizing check valve **170**.

(21) The minimum pressurizing check valve **170** includes, as main components, a cylinder **172** and a piston **174**.

(22) The cylinder **172** is formed into a hollow cylindrical shape, and is provided with a main fuel inflow port (inflow port) **172i** and a reference fuel port **172r** at a first end **172M** and a second end **172R** (indicating a left end and a right end in the drawing, respectively), respectively, in an axial direction of the cylinder. Further, a main fuel outflow port (outflow port) **172o**, which is an opening extending in the axial direction, is provided in a side wall of the cylinder.

(23) The main fuel inflow port **172i** communicates with the outlet (point $P_{sub.M}$) of the metering section **160**, and receives the fuel that has been increased in pressure by the low-pressure pump **110** and the high-pressure pump **120** and has passed through the metering section **160**. In addition, the main fuel outflow port **172o** communicates with the fuel nozzle **320** (through the outlet $P_{sub.O}$ of the fuel supply system **100**), and flows out the fuel received through the main fuel inflow port **172i** to the fuel nozzle **320** of the gas turbine engine **300**. Furthermore, the reference fuel port **172r** communicates with the inlet (point $P_{sub.1}$ (reference pressure acquisition portion)) of the fuel supply system **100** through a reference fuel pipe TR (see FIG. 1) to acquire a reference pressure upstream of the pump (high-pressure pump **120**).

(24) The piston **174** is disposed inside the cylinder **172** and is formed to be movable back and forth in the axial direction of the cylinder **172** in a state where an outer peripheral surface thereof comes into airtight contact with an inner peripheral surface of the cylinder **172**. The piston **174** partitions an internal space of the cylinder **172** into a main fuel chamber CM (first chamber) on a first end **172M** side (left side in the drawing) and a reference fuel chamber CR (second chamber) on a second end **172R** side (right side in the drawing). A spring **176** is disposed in the reference fuel chamber CR to connect the piston **174** to the second end **172R** of the cylinder **172**, and the spring

176 applies a force **F** to the piston **174** in a direction from the second end **172R** toward the first end **172M** of the cylinder **172**.

(25) In the minimum pressurizing check valve **170** configured as described above, depending on fluctuations in the pressure (pressure of the inflowing fuel) in the main fuel chamber **CM** and the pressure (reference pressure) in the reference fuel chamber **CR**, the piston **174** moves to an equilibrium position where a force (restoration force) caused by a differential pressure between the both pressures and the force **F** applied from the spring **176** (restoration force applying member) are balanced, whereby the opening degree (opening area) of the main fuel outflow port **172o** (outflow port) extending in the axial direction is adjusted.

(26) A specific mode of movement of the piston **174** will be described below with reference to FIG. **3**. In the following description, it is assumed that the pressure at the inlet (point **P.sub.I**) of the fuel supply system **100** and the pressure in the reference fuel chamber **CR** are substantially equal to each other and the pressure at the outlet (point **P.sub.M**) of the metering section **160** and the pressure in the main fuel chamber **CM** are substantially equal to each other.

(27) In a state where the electric motor **130** is stopped and the low-pressure pump **110** and the high-pressure pump **120** are not operating, there is no difference between the pressure in the main fuel chamber **CM** and the pressure in the reference fuel chamber **CR**, and the piston **174** is pressed against the first end **172M** of the cylinder **172** by the force **F** of the spring **176** (see FIG. **3A**).

(28) When the electric motor **130** starts up and the low-pressure pump **110** and the high-pressure pump **120** start operating, as loads on these two pumps (for example, the rotation speed of the electric motor **130**) increases, the pressure in the main fuel chamber **CM** gradually increase, and when the force acting on the piston **174** due to the differential pressure from the pressure in the reference fuel chamber **CR** is in a state of overcoming the force **F** of the spring **176**, the piston **174** starts moving toward the second end **172R** of the cylinder **172**. In such a state, the main fuel outflow port **172o** of the cylinder **172** is closed by the piston **174**, but the opening degree of the main fuel outflow port **172o** at this time is expressed as a minus (less than 0%) opening degree for convenience herein.

(29) When the rotation speed of the electric motor **130** increases and the amount of increase in pressure of the low-pressure pump **110** and the high-pressure pump **120** increases, the pressure in the main fuel chamber **CM** increases, and the main fuel outflow port **172o** of the cylinder **172** begins to open (see FIG. **3B**) at a point of time when the differential pressure from the pressure in the reference fuel chamber **CR** exceeds a first threshold $\Delta P_{sub.1}$. In other words, when the above-described differential pressure is equal to the first threshold $\Delta P_{sub.1}$, the opening degree of the main fuel outflow port **172o** is 0%.

(30) When the rotation speed of the electric motor **130** further increases and the amount of increase in pressure of the low-pressure pump **110** and the high-pressure pump **120** further increases, the pressure in the main fuel chamber **CM** further increases, and the main fuel outflow port **172o** of the cylinder **172** becomes fully open (see FIG. **3C**) at a point of time when the differential pressure from the pressure in the reference fuel chamber **CR** reaches a second threshold $\Delta P_{sub.2}$. In other words, when the above-described differential pressure is equal to the second threshold $\Delta P_{sub.2}$, the opening degree of the main fuel outflow port **172o** is 100%.

(31) In this way, when the differential pressure $\Delta P_{sub.T}$ (see FIG. **1**) between the pressure at the inlet (point **P.sub.I**) of the fuel supply system **100** and the pressure at the outlet (point **P.sub.M**) of the metering section **160** is the first threshold $\Delta P_{sub.1}$ or more and the second threshold $\Delta P_{sub.2}$ or less, that is, satisfies Formula (1) below, the minimum pressurizing check valve **170** is configured such that the opening degree of the main fuel outflow port **172o** can be adjusted in a range of 0 to 100% by the piston **174**.

$$(32) \quad P_1 \leq P_T \leq P_2 \quad (1)$$

(33) Here, assuming that due to a change in operating condition of the boost pump **220** of the

aircraft **200**, a discharge pressure of the boost pump **220** can fluctuate within a range of $\pm\Delta P_{\text{sub.B}}$ ($\Delta P_{\text{sub.B}}$: maximum deviation of the discharge pressure) based on a median of fluctuations to be assumed, a situation is considered in which the boost pump **220** is operating with its discharge pressure equal to the median of the fluctuations described above and the fuel supply system **100** is normally operating in a state of satisfying Formula (2) below (that is, the motor **130** is operating at a constant rotation speed).

$$(34) \quad P_1 + P_B \leq P_T \leq P_2 - P_B \quad (2)$$

(35) In this situation, when the discharge pressure of the boost pump **220** is reduced by $\Delta P_{\text{sub.B}}$, the pressure change propagates from the inlet (point P.sub.I) of the fuel supply system **100** to the reference fuel chamber CR through the reference fuel pipe TR and the reference fuel port **172r**. Then, since the pressure in the reference fuel chamber CR temporarily becomes lower by $\Delta P_{\text{sub.B}}$ than the pressure in the main fuel chamber CM, the piston **174** being in the equilibrium position moves toward the second end **172R** (toward the right in the drawing) against the force F of the spring **176**. Thereby, since the opening degree of the main fuel outflow port **172o** increases, the pressure in the main fuel chamber CM located upstream thereof is reduced. As a result, the differential pressure between the pressure in the main fuel chamber CM and the pressure in the reference fuel chamber CR returns to the state before the pressure in the reference fuel chamber CR is reduced, and the piston **174** also returns to the original equilibrium position.

(36) Conversely, when the discharge pressure of the boost pump **220** increases by $\Delta P_{\text{sub.B}}$, the pressure in the reference fuel chamber CR temporarily becomes higher by $\Delta P_{\text{sub.B}}$ than the pressure in the main fuel chamber CM, and thus the piston **174** being in the equilibrium position moves toward the first end **172M** (toward the left in the drawing). Thereby, since the opening degree of the main fuel outflow port **172o** decreases, the pressure in the main fuel chamber CM located upstream thereof increases. As a result, the differential pressure between the pressure in the main fuel chamber CM and the pressure in the reference fuel chamber CR returns to the state before the pressure in the reference fuel chamber CR increases, and the piston **174** also returns to the original equilibrium position.

(37) Here, when the differential pressure $\Delta P_{\text{sub.T}}$ is equal to the maximum value $\Delta P_{\text{sub.2}} - \Delta P_{\text{sub.B}}$ in the range defined by Formula (2) above, the piston **174** is in the equilibrium position where the opening degree of the main fuel outflow port **172o** is slightly less than 100%. When the discharge pressure of the boost pump **220** is reduced by $\Delta P_{\text{sub.B}}$ in this state, the differential pressure $\Delta P_{\text{sub.T}}$ temporarily increases up to $\Delta P_{\text{sub.2}}$, but at this time, the piston **174** moves to a position where the opening degree of the main fuel outflow port **172o** is exactly equal to 100%. In this way, even when the differential pressure $\Delta P_{\text{sub.T}}$ is equal to the maximum value in the range defined by Formula (2) above, the minimum pressurizing check valve **170** has the behavior that the piston **174** returns to its original equilibrium position as described above when the discharge pressure of the boost pump **220** is reduced.

(38) Further, when the differential pressure $\Delta P_{\text{sub.T}}$ is equal to the minimum value $\Delta P_{\text{sub.1}} + \Delta P_{\text{sub.B}}$ in the range defined by Formula (2) above, the piston **174** is in the equilibrium position where the opening degree of the main fuel outflow port **172o** is slightly more than 0%. When the discharge pressure of the boost pump **220** increases by $\Delta P_{\text{sub.B}}$ in this state, the differential pressure $\Delta P_{\text{sub.T}}$ is temporarily reduced up to $\Delta P_{\text{sub.1}}$, but at this time, the piston **174** moves to a position where the opening degree of the main fuel outflow port **172o** is exactly equal to 0%. In this way, even when the differential pressure $\Delta P_{\text{sub.T}}$ is equal to the minimum value in the range defined by Formula (2) above, the minimum pressurizing check valve **170** has the behavior that the piston **174** returns to its original equilibrium position as described above when the discharge pressure of the boost pump **220** increases.

(39) In this way, when the differential pressure $\Delta P_{\text{sub.T}}$ between the pressure at the inlet (point P.sub.I) of the fuel supply system **100** and the pressure at the outlet (point P.sub.M) of the metering

section **160** is within the range defined by Formula (2) above, the minimum pressurizing check valve **170** has the behavior of keeping the differential pressure $\Delta P_{sub.T}$ constant even when the discharge pressure of the boost pump **220** of the aircraft **200** changes.

(40) Therefore, the fuel supply system **100** is configured such that the internal leak model is identified, in the state where Formula (2) above is satisfied, to keep the differential pressure $\Delta P_{sub.T}$ constant during a low fuel flow rate when the internal leak model of the high-pressure pump **120** is identified. This will be described below.

(41) Here, referring again to FIG. **1**, the operation of the fuel supply system **100** will be described.

(42) The fuel supplied from the fuel tank **210** of the aircraft **200** through the boost pump **220** is sequentially increased in pressure by the low-pressure pump **110** and the high-pressure pump **120**, and then flows into the metering section **160**.

(43) During the low fuel flow rate, the pressurizing valve **150** is in a closed state, and all of the fuel flowing into the metering section **160** passes through the orifice **140**. At this time, the signal of the front-back differential pressure $\Delta P_{sub.M}$ of the orifice **140** measured by the differential pressure gauge SDP connected in parallel with the orifice **140** is transmitted to the fuel supply controller **180**, and a fuel flow rate $Q_{sub.M}$ passing through the orifice **140** is calculated based on the relationship between the front-back differential pressure and the fuel flow rate.

(44) On the one hand, the fuel flow rate calculated by the fuel supply controller **180** is compared with a target value of the fuel flow rate input from the controller of the gas turbine engine **300**, and a rotation speed command signal $N_{sub.r}$ is transmitted to the electric motor **130**, which drives the low-pressure pump **110** and the high-pressure pump **120**, to make the deviation between both flow rates zero. On the other hand, the fuel supply controller **180** uses the calculated fuel flow rate $Q_{sub.M}$ to identify the internal leak model of the high-pressure pump **120**. This will be described below.

(45) The internal leak model of the high-pressure pump **120** is a model for calculating an internal leak area $A_{sub.leak}$ of the high-pressure pump **120** using the fuel flow rate $Q_{sub.M}$ described above, and is expressed by Formula (3) below.

$$(46) [\text{Equation1}] \quad A_{leak} = \frac{V_{th} \cdot \text{Math. } N_H - Q_M}{Cd} \sqrt{\frac{2}{P_H}} \quad (3)$$

(47) Where, $V_{sub.th}$ represents a theoretical discharge volume (constant) of the high-pressure pump **120**, $N_{sub.H}$ represents the rotation speed of the high-pressure pump **120** (calculated from the rotation speed $N_{sub.m}$ measured by the rotation speed sensor SN attached to the electric motor **130**), Cd represents a flowrate coefficient (constant) of the high-pressure pump **120**, ρ represents density of the fuel at the inlet of the high-pressure pump **120** (calculated from the fuel temperature T at the inlet of the high-pressure pump **120** measured by the temperature sensor ST), and $\Delta P_{sub.H}$ represents the front-back differential pressure of the high-pressure pump **120**.

(48) The internal leak area $A_{sub.leak}$ of the high-pressure pump **120** calculated based on the internal leak model is then used to calculate volumetric efficiency η_v of the high-pressure pump **120** using Formula (4) below.

$$(49) [\text{Equation2}] \quad \eta_v = \frac{Cd \cdot \text{Math. } A_{leak} \sqrt{\frac{2}{P_H}}}{V_{th} \cdot \text{Math. } N_H} \quad (4)$$

(50) The volumetric efficiency η_v of the high-pressure pump **120** calculated in this way is used to estimate the fuel flow rate (discharge flow rate of the high-pressure pump **120**) $Q_{sub.H}$ using Formula (5) below at the time of high fuel flow rate (that is, when the pressurizing valve **150** is in an open state).

$$(51) \quad Q_H = \eta_v \cdot \text{Math. } V_{th} \cdot \text{Math. } N_H \quad (5)$$

(52) In the identification of the internal leak model of the high-pressure pump **120** described above, the front-back differential pressure $\Delta P_{sub.H}$ of the high-pressure pump **120** is calculated by Formula (6) below.

$$(53) \quad P_H = P_T - (P_L + P_M) \quad (6)$$

(54) Here, $\Delta P_{\text{sub.T}}$ represents the differential pressure between the pressure at the inlet (point P.sub.I) of the fuel supply system **100** and the pressure at the outlet (point P.sub.M) of the metering section **160**, as described above, $\Delta P_{\text{sub.L}}$ represents the amount of pressure increase in the low-pressure pump **110**, and $\Delta P_{\text{sub.M}}$ represents the front-back differential pressure of the metering section **160** (orifice **140**) (see FIG. 1).

(55) In these items, $\Delta P_{\text{sub.L}}$ is estimated based on Formula (7) below using the rotation speed N.sub.m of the electric motor **130** that drives the low-pressure pump **110**. Here, $f_{\text{sub.L}}$ is a function of N.sub.m, and is determined in advance by tests or the like.

$$(56) \quad P_L = f_L(N_m) \quad (7)$$

(57) Further, $\Delta P_{\text{sub.M}}$ is measured by the differential pressure gauge SDP connected in parallel with the orifice **140**.

(58) On the other hand, according to the fuel supply system disclosed in Patent Document 1, $\Delta P_{\text{sub.T}}$ can be calculated as a difference between the pressure at the inlet of the fuel supply system and the pressure at the outlet of the metering section which are measured by pressure sensors, respectively.

(59) However, since these pressure sensors are not provided in the fuel supply system **100**, $\Delta P_{\text{sub.T}}$ is estimated based on Formula (8) below using the rotation speed N.sub.m of the electric motor **130** that drives the low-pressure pump **110** and the high-pressure pump **120**, similarly to $\Delta P_{\text{sub.L}}$. Here, $f_{\text{sub.T}}$ is a function of N.sub.m, and is determined in advance by tests or the like.

$$(60) \quad P_T = f_T(N_m) \quad (8)$$

(61) In the fuel supply system **100**, the behavior of the minimum pressurizing check valve **170** described above is used in order to keep the differential pressure $\Delta P_{\text{sub.T}}$ in Formula (6) constant such that the front-back differential pressure $\Delta P_{\text{sub.H}}$ of the high-pressure pump **120** can be calculated with high accuracy using Formula (6) above when the identification of the internal leak model of the high-pressure pump **120** is performed.

(62) As described above, when the differential pressure $\Delta P_{\text{sub.T}}$ is within the range defined by Formula (2) above, the minimum pressurizing check valve **170** has the behavior to keep the differential pressure $\Delta P_{\text{sub.T}}$ constant even when the discharge pressure of the boost pump **220** of the aircraft **200** changes.

(63) Therefore, the fuel supply system **100** is configured such that the identification of the internal leak model of the high-pressure pump **120** can be performed when the rotation speed N.sub.m of the electric motor **130** satisfies Formula (9) below.

$$(64) \quad N_1 \leq N_m \leq N_2 \quad (9)$$

(65) Here, N.sub.1 represents the rotation speed (lower threshold) of the electric motor **130** at which the differential pressure $\Delta P_{\text{sub.T}}$ becomes equal to the minimum value $\Delta P_{\text{sub.1}} + \Delta P_{\text{sub.B}}$ in the range defined by Formula (2) above, and N.sub.2 represents the rotation speed (intermediate threshold) of the electric motor **130** at which the differential pressure $\Delta P_{\text{sub.T}}$ becomes equal to the maximum value $\Delta P_{\text{sub.2}} - \Delta P_{\text{sub.B}}$ in the range defined by Formula (2) above.

(66) In order to realize the above-described configuration, the minimum pressurizing check valve **170** is configured such that the opening degree of the main fuel outflow port **172o** can be variably adjusted in the range of 0 to 100% in the rotation speed range defined by Formula (9) above.

(67) Specifically, the minimum pressurizing check valve **170** is designed to satisfy conditions (A) and (B) below (specifically, a dimension of each of the cylinder **172** and the piston **174** and a spring constant of the spring **176** are set). (A) When the rotation speed N.sub.m of the electric motor **130** is equal to the minimum value N.sub.1 in the range defined by Formula (9) above, the opening degree of the main fuel outflow port **172o** is exactly 0% in the case where the pressure in the reference fuel chamber CR increases by $\Delta P_{\text{sub.B}}$. (B) When the rotation speed N.sub.m of the electric motor **130** is equal to the maximum value N.sub.2 in the range defined by Formula (9) above, the opening degree of the main fuel outflow port **172o** is exactly 100% in the case where the

pressure in the reference fuel chamber CR is reduced by $\Delta P_{\text{sub.B}}$.

(68) When the rotation speed $N_{\text{sub.m}}$ of the electric motor **130** reaches a rotation speed $N_{\text{sub.3}}$ which is larger than the rotation speed $N_{\text{sub.2}}$ described above, the front-back differential pressure of the orifice **140** increases as the fuel flow rate increases, and the differential pressure reaches the threshold inherent to the pressurizing valve **150** which is a check valve, whereby the pressurizing valve **150** begins to open.

(69) In the state where the pressurizing valve **150** is open as described above, the fuel flow rate $Q_{\text{sub.M}}$ passing through the orifice **140** no longer matches the total fuel flow rate supplied from the fuel supply system **100** to the fuel nozzle **320** of the gas turbine engine **300**.

(70) For this reason, in the state where the rotation speed $N_{\text{sub.m}}$ of the electric motor **130** is equal to or higher than the rotation speed $N_{\text{sub.3}}$ (upper threshold), using the internal leak model of the high-pressure pump **120** identified as described above, the fuel flow rate (discharge flow rate of the high-pressure pump **120**) $Q_{\text{sub.H}}$ is estimated based on Formulas (4) and (5) above, and the rotation speed of the electric motor **130** is controlled such that the deviation between the estimated value and the target value becomes zero.

(71) In such a state, the piston **174** of the minimum pressurizing check valve **170** is pressed against the second end **172R** of the cylinder **172**, and the main fuel outflow port **172o** of the cylinder **172** is fully open (see FIG. 3D). In the description, the opening degree of the main fuel outflow port **172o** in this state is regarded as an opening degree larger than 100% for convenience.

(72) The embodiment has been described above in which the reference fuel chamber CR of the minimum pressurizing check valve **170** communicates with the inlet (point $P_{\text{sub.I}}$) of the fuel supply system **100** through the reference fuel port **172r** and the reference fuel pipe TR, but the reference fuel chamber CR of the minimum pressurizing check valve **170** may communicate with the inlet (point $P_{\text{sub.L}}$) of the high-pressure pump **120**.

(73) Further, the embodiment has been described above in which the temperature sensor ST is provided at the point $P_{\text{sub.L}}$ on the fuel flow passage between the low-pressure pump **110** and the high-pressure pump **120**, but the temperature sensor ST may be provided at the inlet (point $P_{\text{sub.I}}$) of the fuel supply system **100**.

(74) Furthermore, the embodiment has been described above in which the opening degree of the main fuel outflow port **172o** of the minimum pressurizing check valve **170** is 0% when the differential pressure between the pressure in the main fuel chamber CM and the pressure in the reference fuel chamber CR is equal to the first threshold $\Delta P_{\text{sub.1}}$, and is 100% when the differential pressure therebetween is equal to the second threshold $\Delta P_{\text{sub.2}}$, but these opening degrees may be 0% or more, and may be 100% or less, respectively.

(75) As described above, the fuel supply system **100** of the aircraft gas turbine engine according to the embodiment of the present disclosure employs the minimum pressurizing check valve **170** having the main fuel outflow port **172o** configured such that the opening degree thereof is in the range of 0 to 100% even when the discharge pressure of the boost pump **220** of the aircraft **200** changes while the internal leak model of the high-pressure pump **120** is identified, and thus it is possible to keep the differential pressure $\Delta P_{\text{sub.T}}$ between the system inlet (point $P_{\text{sub.I}}$) and the minimum pressurizing check valve **170** constant during this time, making it possible to identify the internal leak model described above with high accuracy.

(76) (Aspects of Present Disclosure)

(77) A fuel supply system according to a first aspect of the present disclosure includes: a gear pump that increases a pressure of fuel flowing in from a system inlet and discharges the fuel; a metering section that is disposed downstream of the gear pump and includes an orifice and a check valve connected in parallel with each other; a minimum pressurizing check valve that is disposed between the metering section and a system outlet; and a controller, wherein the check valve closes when a rotation speed of the gear pump is less than an upper threshold, and opens when the rotation speed of the gear pump is equal to or more than the upper threshold, the minimum pressurizing

check valve includes a cylinder and a piston capable of being movable back and forth in the cylinder, the cylinder has an internal space partitioned by the piston into a first chamber that communicates with an outlet of the metering section through an inflow port and a second chamber that communicates with the system inlet, the piston receives a force in a direction toward the first chamber from a spring disposed in the second chamber, a side wall of the cylinder is provided with an outflow port that is an opening through which the inflow port and the system outlet communicate with each other, an opening degree of the outflow port is adjusted as the piston moves back and forth, and the minimum pressurizing check valve keeps a differential pressure between the system inlet and the outlet of the metering section constant when the rotation speed of the gear pump is equal to or more than a lower threshold and is equal to or less than an intermediate threshold, the lower threshold being smaller than the upper threshold, the intermediate threshold being more than the lower threshold and smaller than the upper threshold.

(78) In a fuel supply system according to a second aspect of the present disclosure, the minimum pressurizing check valve is capable of variably adjusting an opening degree of the outflow port in a range of 0 to 100% when the rotation speed of the gear pump is equal to or more than the lower threshold and is equal to or less than the intermediate threshold.

(79) In a fuel supply system according to a third aspect of the present disclosure, the opening degree of the outflow port is 0% or more when the rotation speed of the gear pump is equal to the lower threshold and a pressure at the system inlet is higher by a predetermined maximum deviation than a median of fluctuations to be assumed, and the opening degree of the outflow port is 100% or less when the rotation speed of the gear pump is equal to the intermediate threshold and the pressure at the system inlet is lower by the maximum deviation than the median of the fluctuations.

(80) A fuel supply method according to a first aspect of the present disclosure is performed using the fuel supply system according to any one of the first to third aspects; in the fuel supply method, when the rotation speed of the gear pump is equal to or more than the lower threshold and is less than the upper threshold, the controller controls the rotation speed of the gear pump based on a fuel flow rate to be measured using the orifice, when the rotation speed of the gear pump is equal to or more than the lower threshold and is equal to or less than the intermediate threshold, the controller identifies an internal leak model of the gear pump based on the fuel flow rate to be measured using the orifice, and when the rotation speed of the gear pump is equal to or more than the upper threshold, the controller controls the rotation speed of the gear pump based on a fuel flow rate to be estimated using the rotation speed of the gear pump and the internal leak model of the gear pump that has been identified.

(81) A fuel supply system for a gas turbine according to a first aspect of the present disclosure includes: a pump; a reference pressure acquisition portion that acquires a reference pressure upstream of the pump; a branch provided downstream of the pump; a check valve provided downstream of the branch; a parallel flow passage provided in parallel with the check valve through the branch; a flowmeter that measures a flow rate based on the parallel flow passage; and a minimum pressurizing check valve provided downstream of the parallel flow passage, wherein the minimum pressurizing check valve includes an outflow port of which opening area increases as a difference between a pressure of inflowing fuel and the reference pressure increases, and the check valve switches to a state where fuel is capable of flowing through the outflow port as a load of the pump increases.

(82) In a fuel supply system for a gas turbine according to a second aspect of the present disclosure, the minimum pressurizing check valve includes: a cylinder provided with the outflow port; a piston movable within the cylinder; and a first chamber and a second chamber in the cylinder partitioned by the piston, wherein the second chamber is connected to the reference pressure acquisition portion to acquire a reference pressure, fuel being increased in pressure through the pump flows into the first chamber, the piston is attached with a member that applies a restoration force toward the first chamber from the second chamber, and the piston is movable to a position covering at least

a part of the outflow port.

(83) In a fuel supply system for a gas turbine according to a third aspect of the present disclosure, the outflow port is covered and closed by the piston when the pump is not driven.

(84) A leak model identification method according to a first aspect of the present disclosure is performed using the fuel supply system for a gas turbine according to any one of the first to third aspects, the method including: a step of driving the pump; and a step of identifying a leak model, based on the flow rate and the load of the pump under a load in which the pump does not allow fuel to flow through the check valve.

EXPLANATION OF REFERENCE SIGNS

(85) **100** fuel supply system **120** high-pressure pump (gear pump) **140** orifice **150** pressurizing valve (check valve) **160** metering section **170** minimum pressurizing check valve **172** cylinder **172o** main fuel outflow port (outflow port) **174** piston **176** spring **180** controller CM main fuel chamber (first chamber) CR reference fuel chamber (second chamber) N.sub.1 lower threshold of rotation speed N.sub.2 intermediate threshold of rotation speed N.sub.3 upper threshold of rotation speed P.sub.I inlet of fuel supply system P.sub.M outlet of metering section

Claims

1. A fuel supply system comprising: a gear pump that increases a pressure of fuel flowing in from a system inlet and discharges the fuel; a metering section that is disposed downstream of the gear pump and includes an orifice and a check valve connected in parallel with each other; a minimum pressurizing check valve that is disposed between the metering section and a system outlet; and a controller, wherein the check valve closes when a rotation speed of the gear pump is less than an upper threshold, and opens when the rotation speed of the gear pump is equal to or more than the upper threshold, the minimum pressurizing check valve includes a cylinder and a piston movable back and forth in the cylinder, the cylinder has an internal space partitioned by the piston into a first chamber that communicates with an outlet of the metering section through an inflow port and a second chamber that communicates with the system inlet, the piston receiving a force in a direction toward the first chamber from a spring disposed in the second chamber, a side wall of the cylinder is provided with an outflow port that is an opening through which the inflow port and the system outlet communicate with each other, an opening degree of the outflow port is adjusted as the piston moves back and forth, and the minimum pressurizing check valve keeps a differential pressure between the system inlet and the outlet of the metering section constant and variably adjusts an opening degree of the outflow port in a range of 0 to 100% when the rotation speed of the gear pump is equal to or more than a lower threshold and is equal to or less than an intermediate threshold, the lower threshold being smaller than the upper threshold, the intermediate threshold being more than the lower threshold and smaller than the upper threshold, the opening degree of the outflow port is 0% or more when the rotation speed of the gear pump is equal to the lower threshold and a pressure at the system inlet is higher by a predetermined maximum deviation than a median of fluctuations to be assumed, and the opening degree of the outflow port is equal to 100% or less when the rotation speed of the gear pump is equal to the intermediate threshold and the pressure at the system inlet is lower by the predetermined maximum deviation than the median of the fluctuations.

2. A fuel supply method using the fuel supply system according to claim 1, wherein when the rotation speed of the gear pump is equal to or more than the lower threshold and is less than the upper threshold, the controller controls the rotation speed of the gear pump based on a fuel flow rate to be measured using an orifice flowmeter including a differential pressure gauge connected in parallel with the orifice, when the rotation speed of the gear pump is equal to or more than the lower threshold and is equal to or less than the intermediate threshold, the controller identifies an internal leak model of the gear pump based on the fuel flow rate to be measured using the orifice

flowmeter, and when the rotation speed of the gear pump is equal to or more than the upper threshold, the controller controls the rotation speed of the gear pump based on a fuel flow rate to be estimated using the rotation speed of the gear pump and the internal leak model of the gear pump that has been identified.
